

SYSTEM DESIGN

Analytics and planning on Fiori

System design, features, test results, and how it compares with SAP
Analytics Cloud

One system for plan and data

1

The need

Planning and analysis on S/4HANA data today means a second product with its own license, connections and security.

2

The question

How much of the SAC working model can run inside the ERP landscape, under the ERP's own authorizations?

3

The answer so far

Stories, models, planning with locking and rules, data and multi actions, a calendar, CDS sources and sharing, tested end to end on a real system with two users.

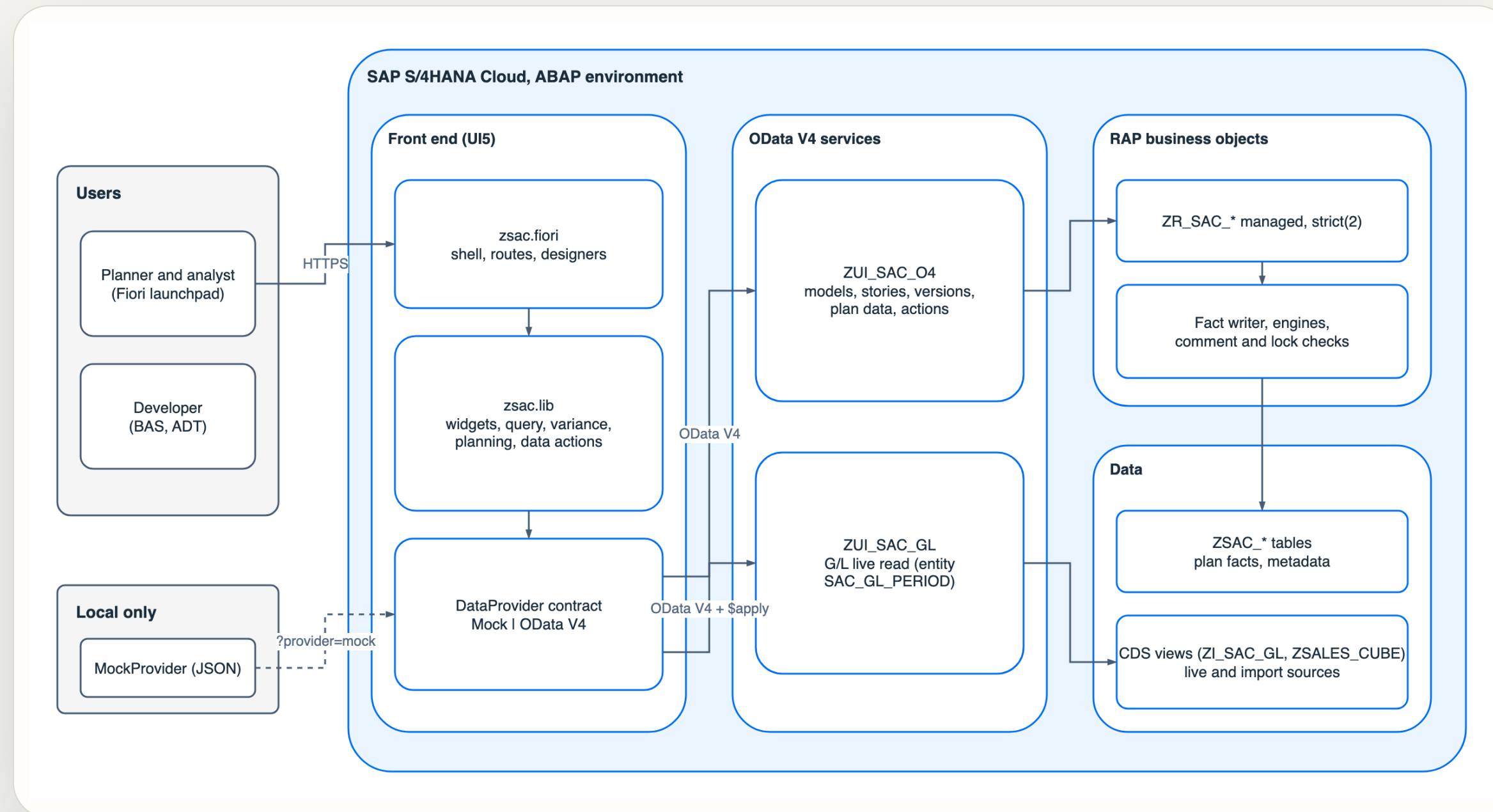
Build, analyse and see

AREA	CAPABILITY
Stories	Pages on a grid; eleven chart types, KPI, tables, input controls, text, images, web pages, feeds, comments, buttons
Data Analyser	Free analysis with rows, columns and filters; calculated measures; CSV and Excel export
Models	Measures with exception aggregation, hierarchies, Date hierarchy, currencies, CSV and CDS sources
Insight	Variance explainer, value driver tree, Compass simulation (Monte Carlo)
Navigation	A floating button opens a menu in the order of the work; usable on a phone
Story layout	Resize from every edge, arrow keys, Tidy up, saved views of tables

Plan, automate and share

AREA	CAPABILITY
Planning	Editable table, spreading, formulas, copy and paste, versions, publish, history, comments
Control	Data Locking by region, validation rules (error or warning), audit log
Data actions	Seven step types: copy, allocation, formula, currency conversion, scale, delete, embedded
Multi actions	Eleven step types: data action, publish, versions, locking, import, forecast, API, PaPM, comments, copy to model, CDS import
Calendar	Processes and tasks, dependencies, approvals, reminders, Gantt, month, week, day and list views
Sharing	Owner, view, edit and everyone, enforced by the server for stories, models, actions and events

The big picture



Three design principles

Modular

The library `zsac.lib` serves any Fiori app. Widgets and data sources are plug-ins registered in one place.

Metadata is data

Models, stories, actions and files are rows, edited by the same screens that show them.

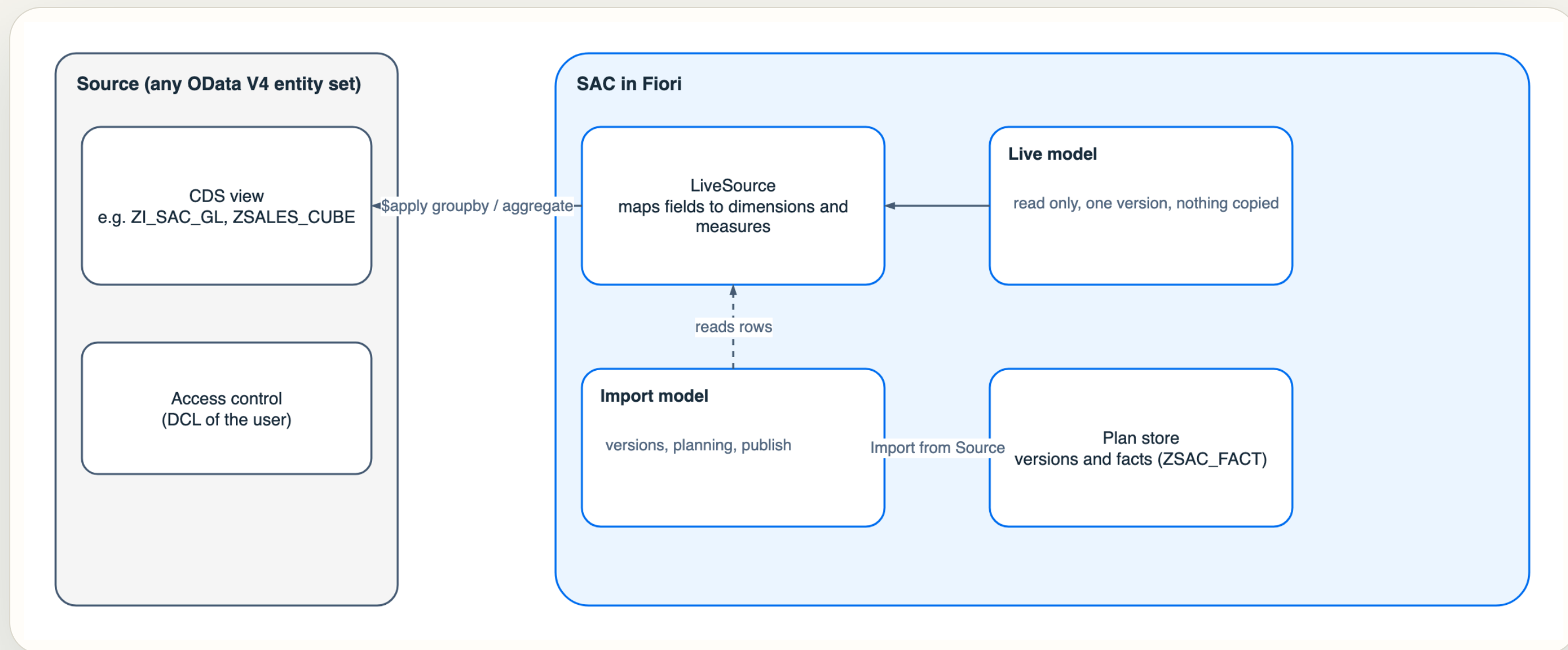
One provider contract

Mock for development and tests, OData V4 for the backend, CDS views for outside data.

DATA

Live and import sources

The service aggregates with \$apply; if it cannot, the app reads the rows and aggregates itself.



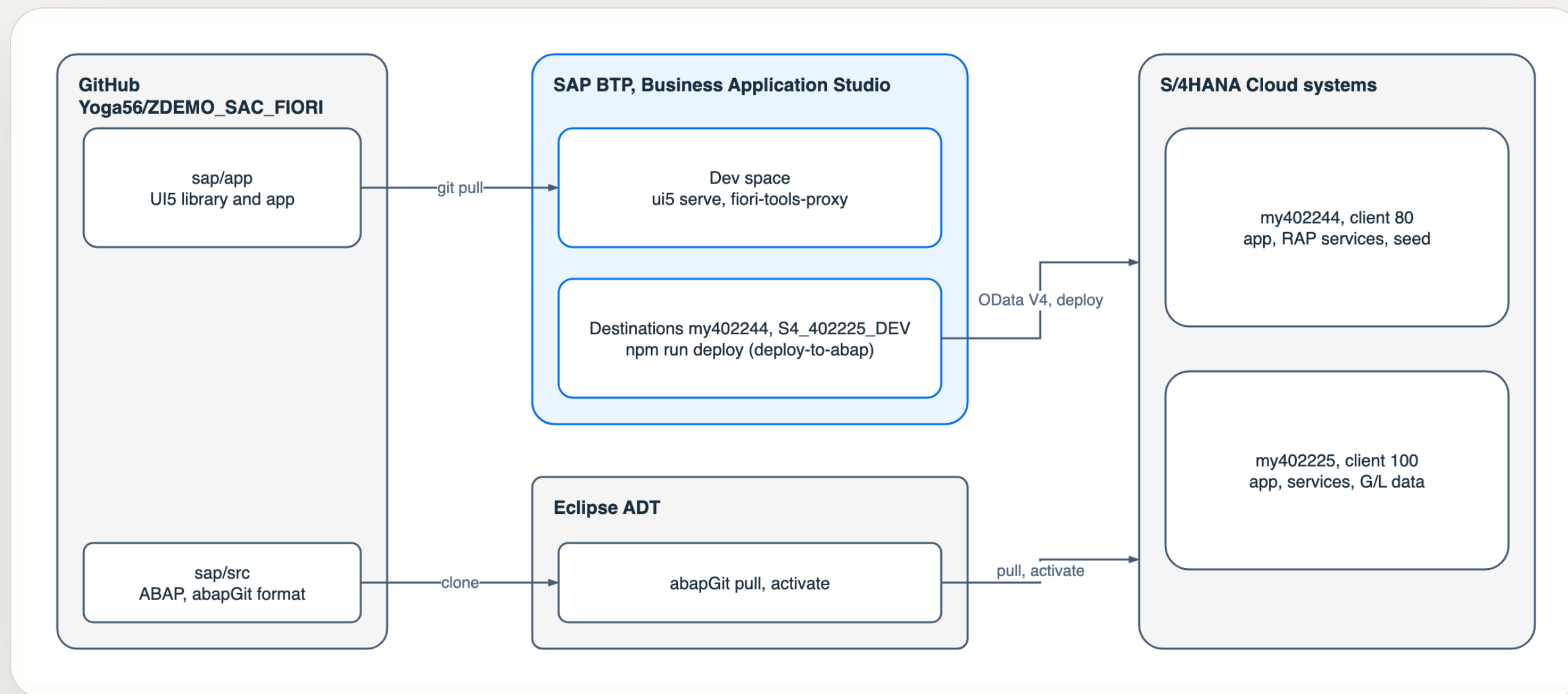
ABAP RAP at a glance

PART	DESIGN
Tables	ZSAC_* tables for files, stories, models, versions, facts, actions, runs, shares and calendar events
Business objects	Managed, strict(2); one request writes a root and its children
Actions	Version: create private, publish, revert. Facts: bulk write and delete
Server rules	Locked versions refuse writes; model structure is checked on save; only the owner shares
Access control	Owner, shared with me and everyone on every projection; the server refuses what the client hides
Services	ZUI_SAC_O4 for the app, ZUI_SAC_BIZ with business views (G/L, sales) as live sources

DELIVERY

Landscape and delivery

The same code runs on mock data, in Business Application Studio and deployed to the system.



Security and operations

In place

Calls run under the user's own session; access controls apply to every read, also on the projections.

In place

Owner, view, edit and everyone for stories, models, actions and events; checked with two users.

In place

Locked versions, locking regions and validation rules refuse writes on the server and in every engine.

In place

Every action run is recorded: status, counts, duration, user.

Open

No row level security on plan data; actions run in the browser, not as a server job.

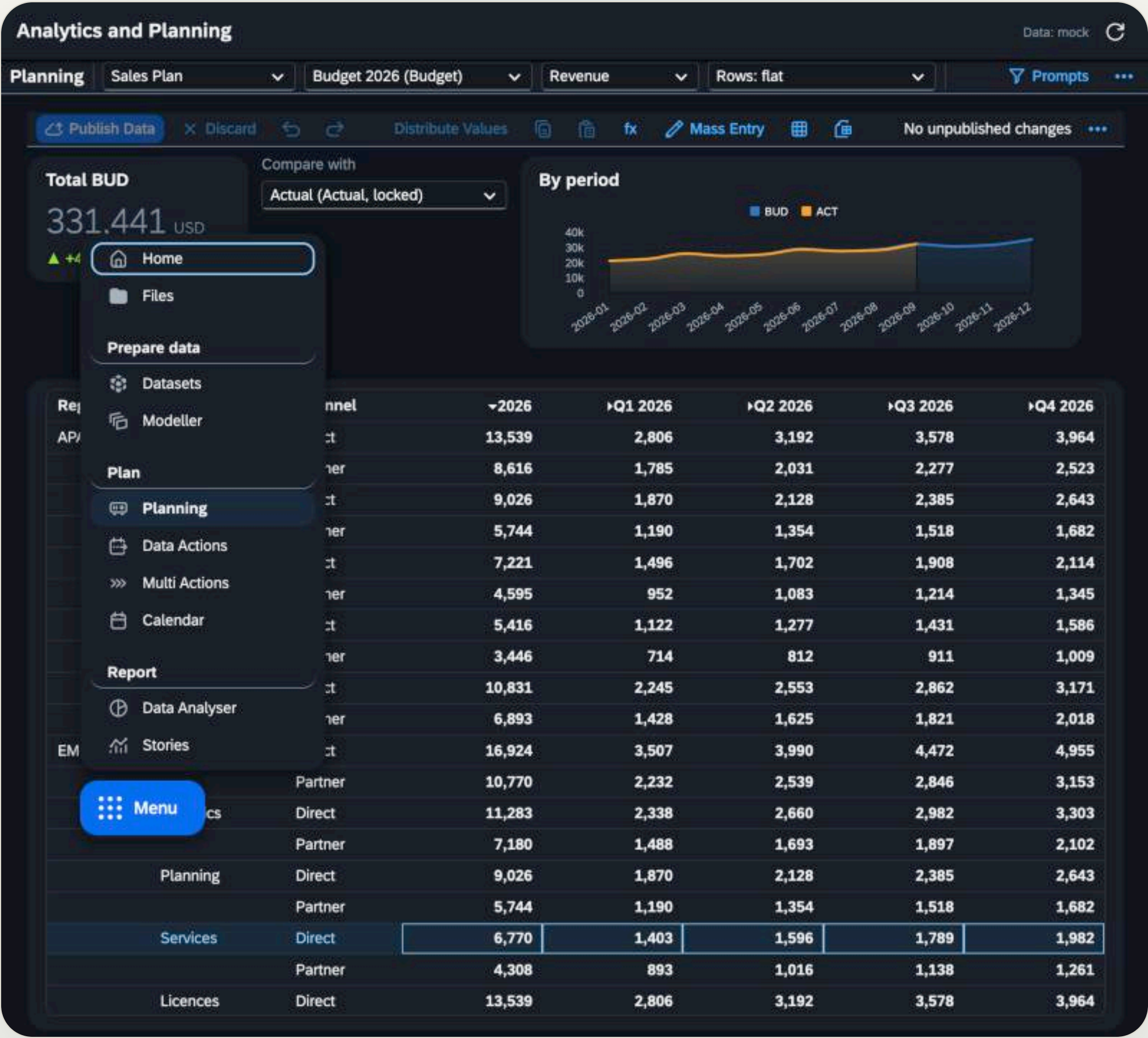
Open

Versions and cell comments are not owned; content moves only through abapGit and the seed class.

NAVIGATION

The floating menu

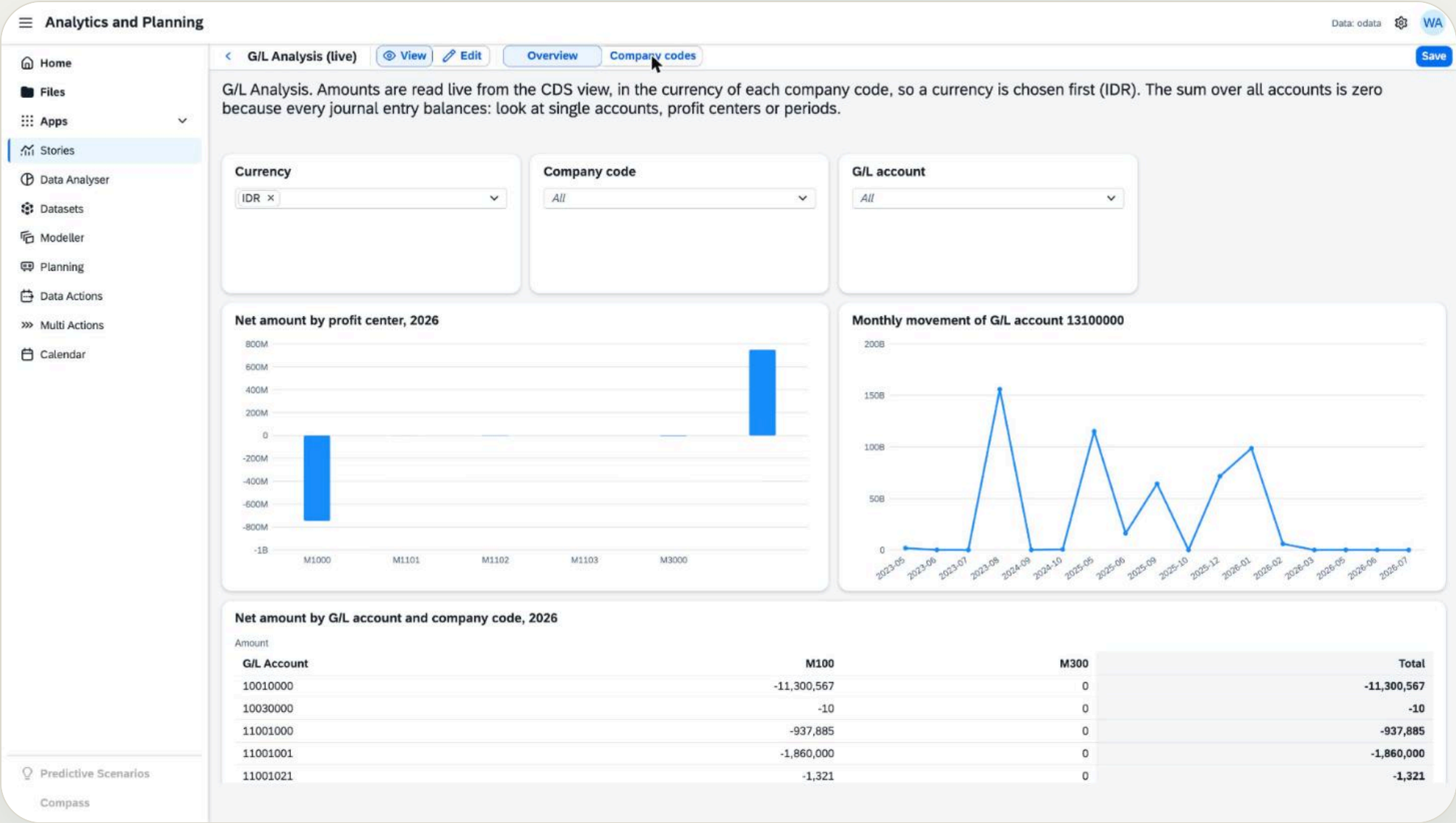
No side bar. A button opens the menu above or below it, wherever there is room; it can be dragged out of the way. Pages are grouped in the order of the work: prepare data, plan, report.



REAL SYSTEM · G/L

Live G/L analysis

A story on the G/L CDS view with 1,092 rows, read live from the S/4HANA Cloud system.

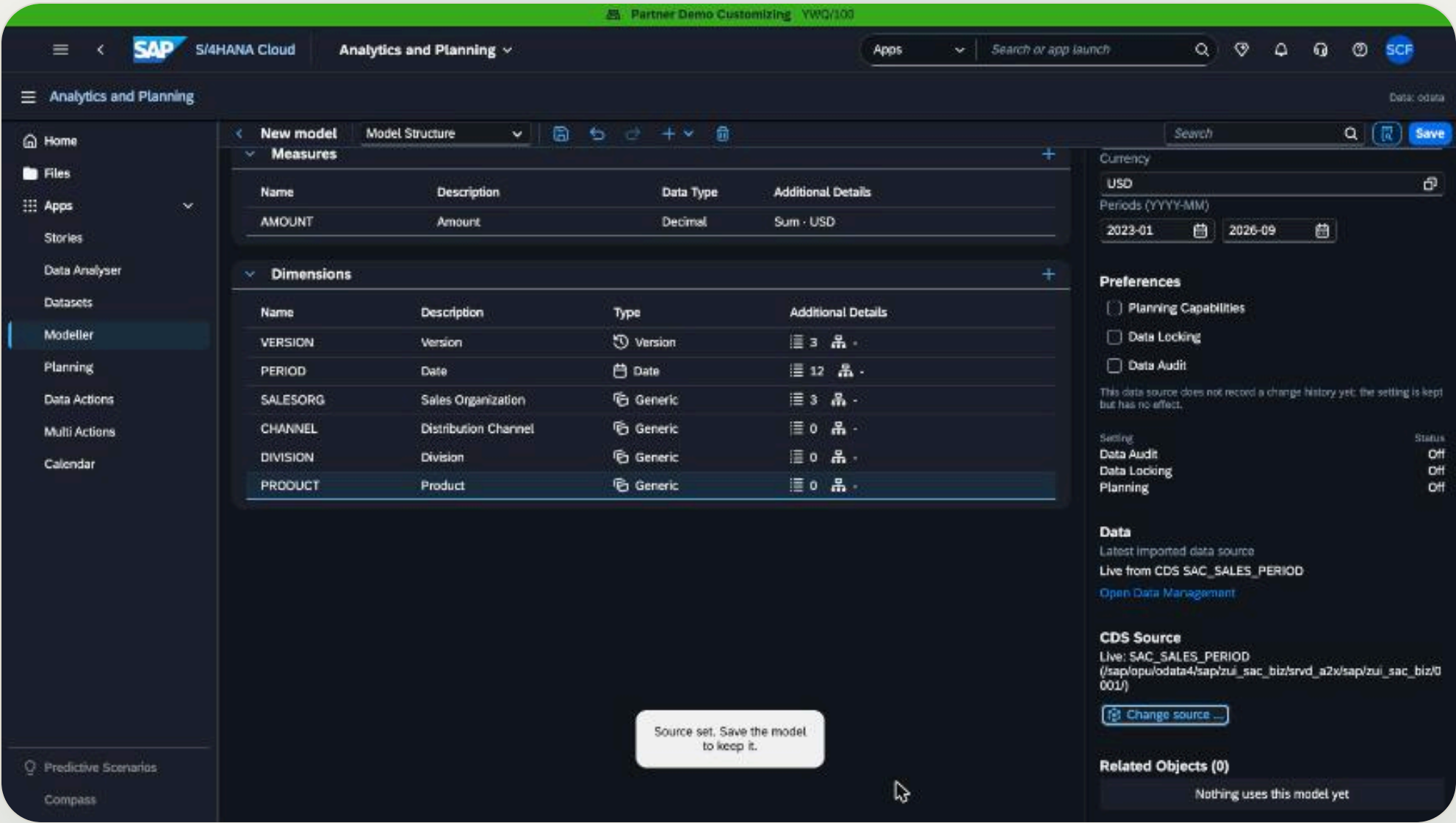


Story G/L Analysis, page 1, on system my402225

REAL SYSTEM · G/L

The model on a CDS view

The live model maps fields of the service to dimensions and measures. The source panel shows the service, entity and mapping.

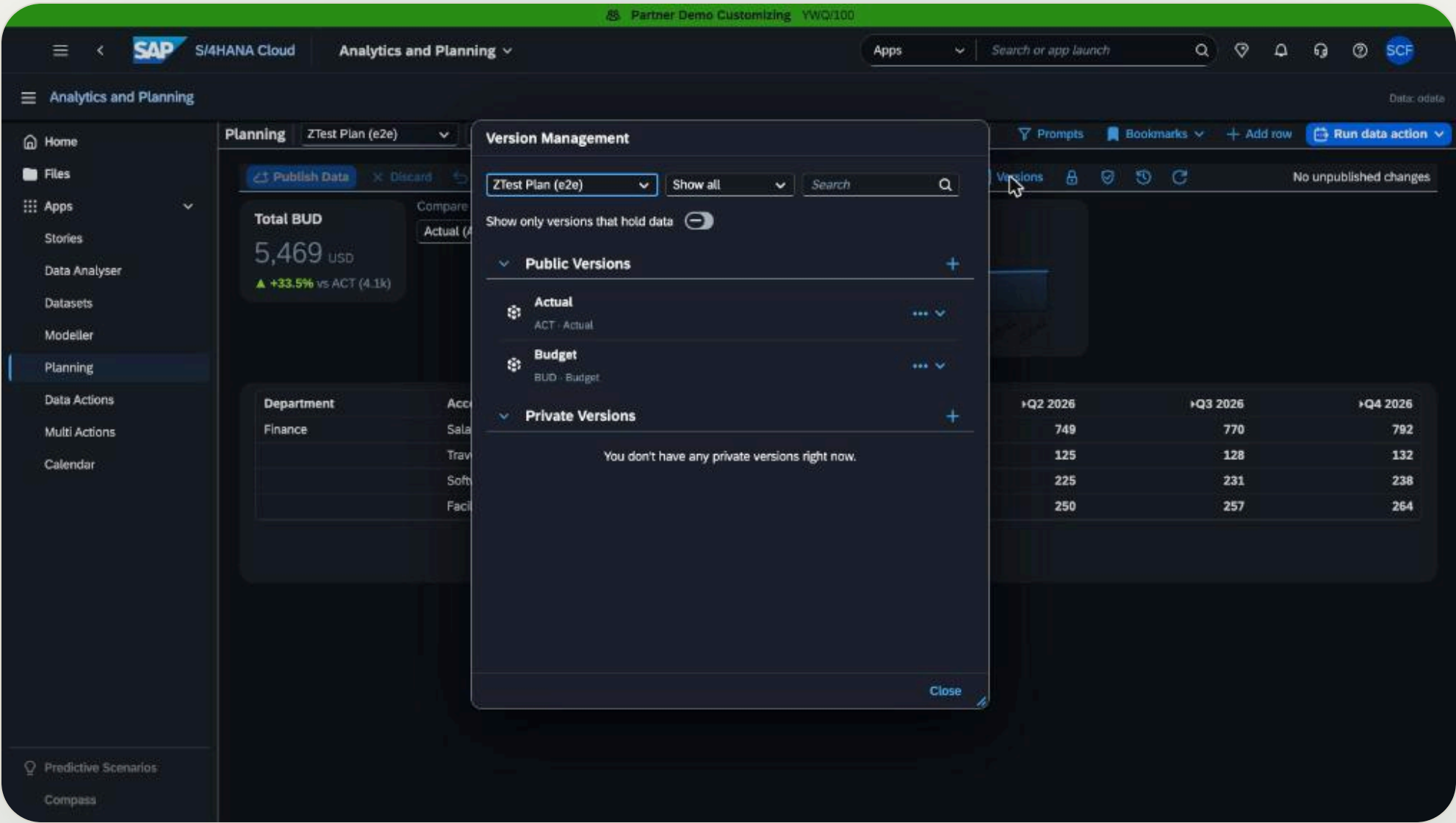


Modeller: live CDS source on the sales view

REAL SYSTEM · PLANNING

Versions and edits

A planner works in a private copy of the budget. Nothing is public until it is published.

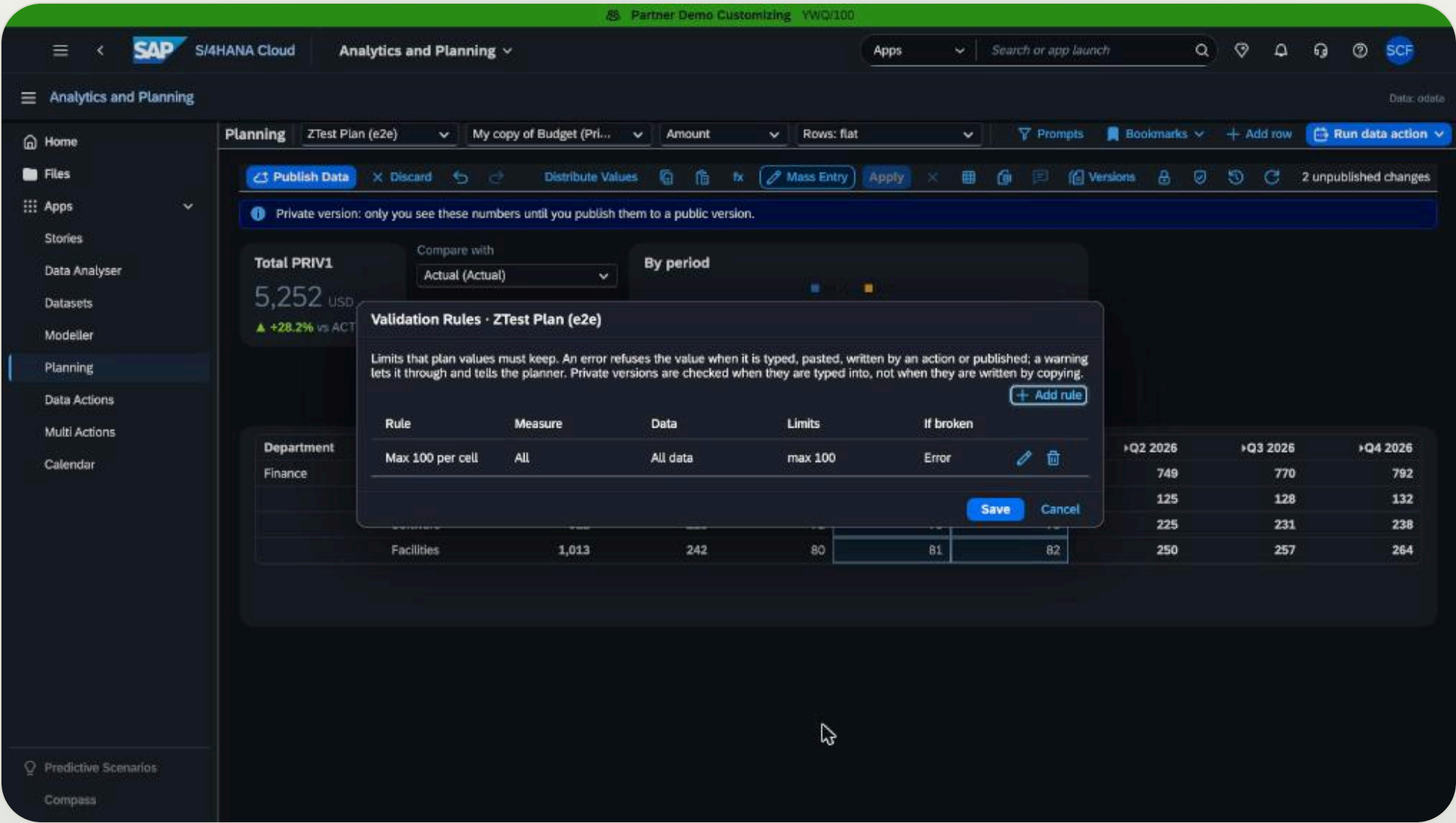


Planning: versions dialog

REAL SYSTEM · PLANNING

Validation rules

A limit for plan values. An error refuses the value; a warning lets it through and says so.

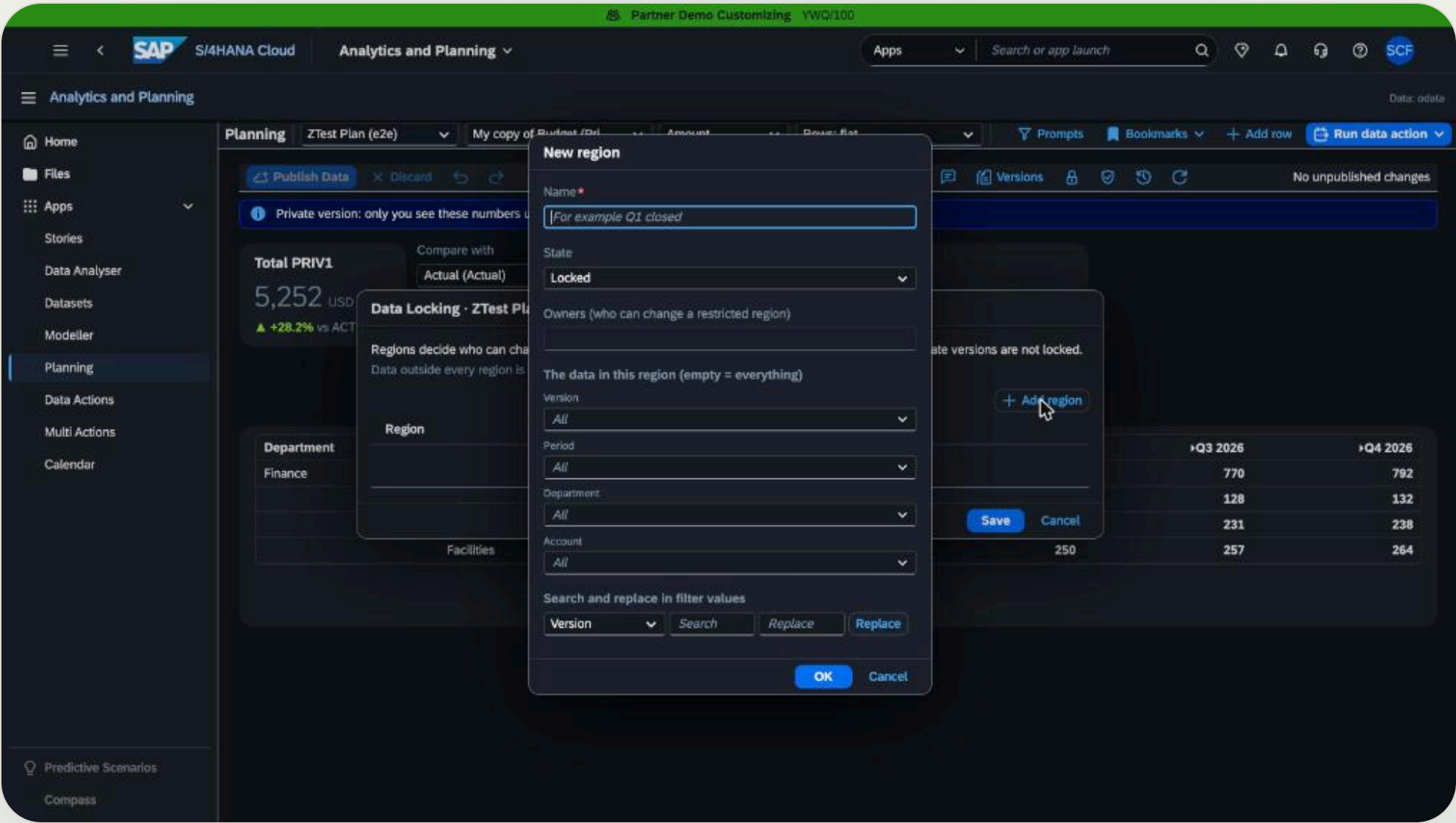


Rule: max 100 per cell, error

REAL SYSTEM · PLANNING

Data locking

A region locks part of the data. The locked cells are marked in the table.

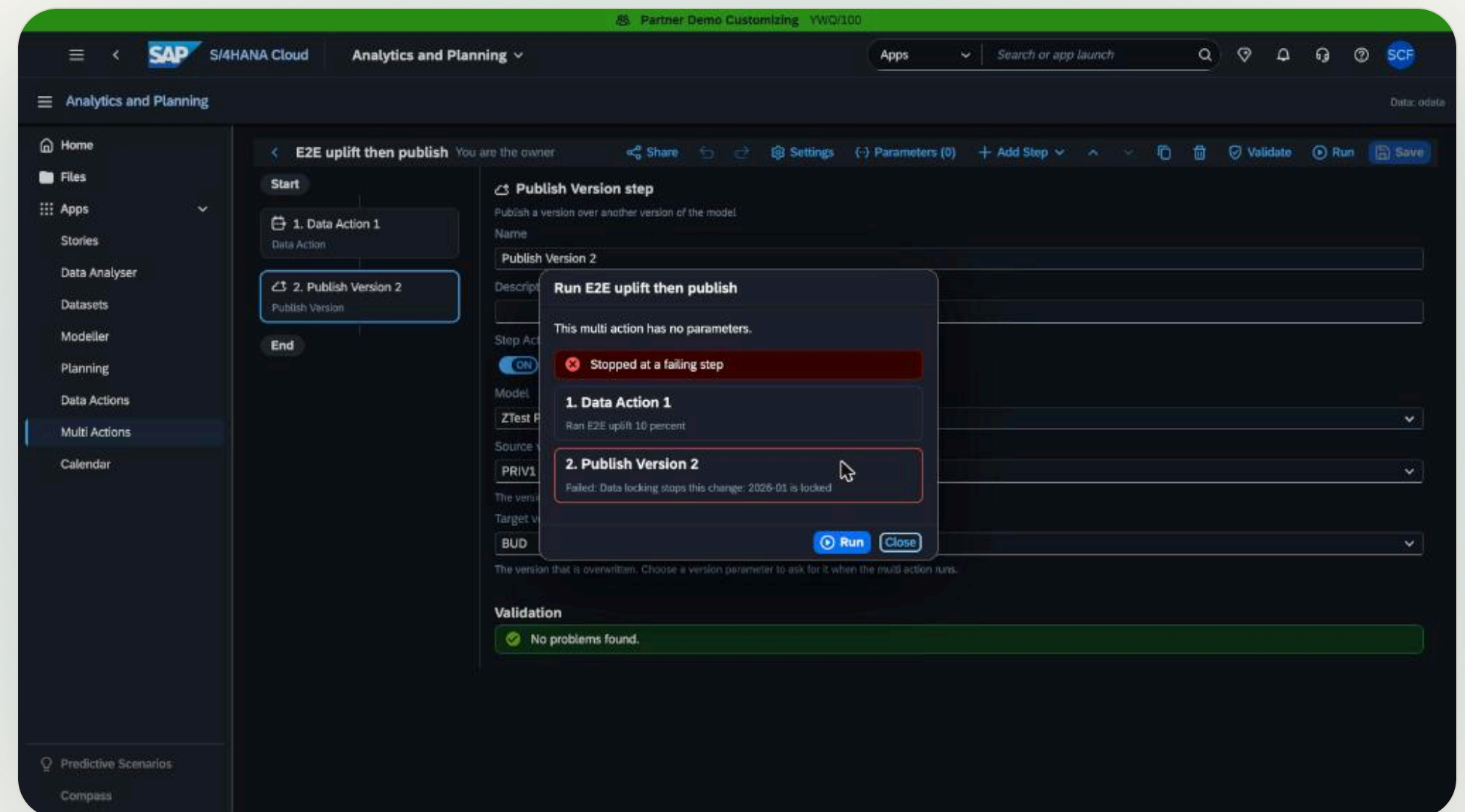


Data Locking dialog

REAL SYSTEM · ACTIONS

A lock stops the run

The multi action stops at the failing publish step and says why; the earlier step stays done.

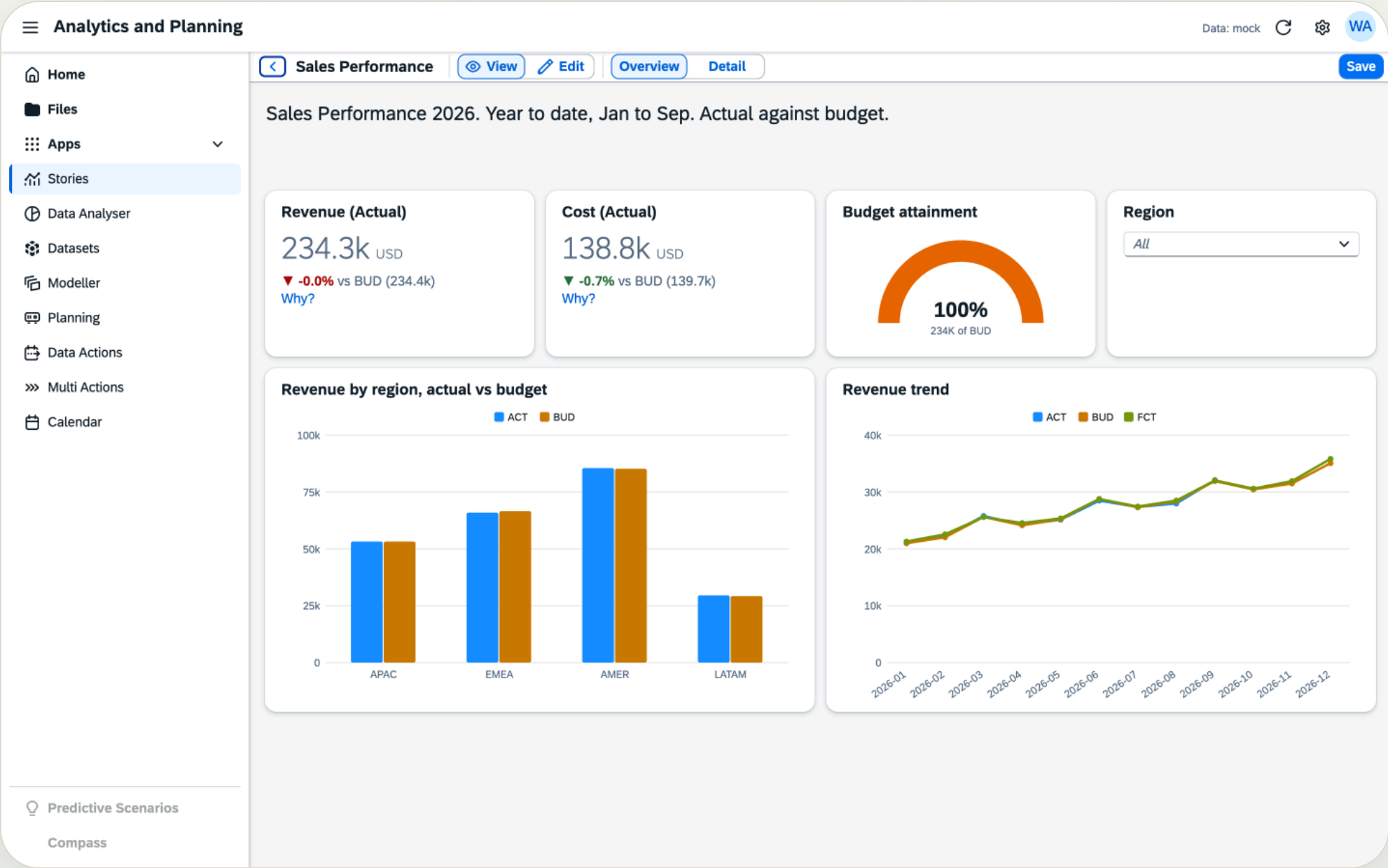


Multi action: data locking stops the publish

SAMPLE DATA

Stories and charts

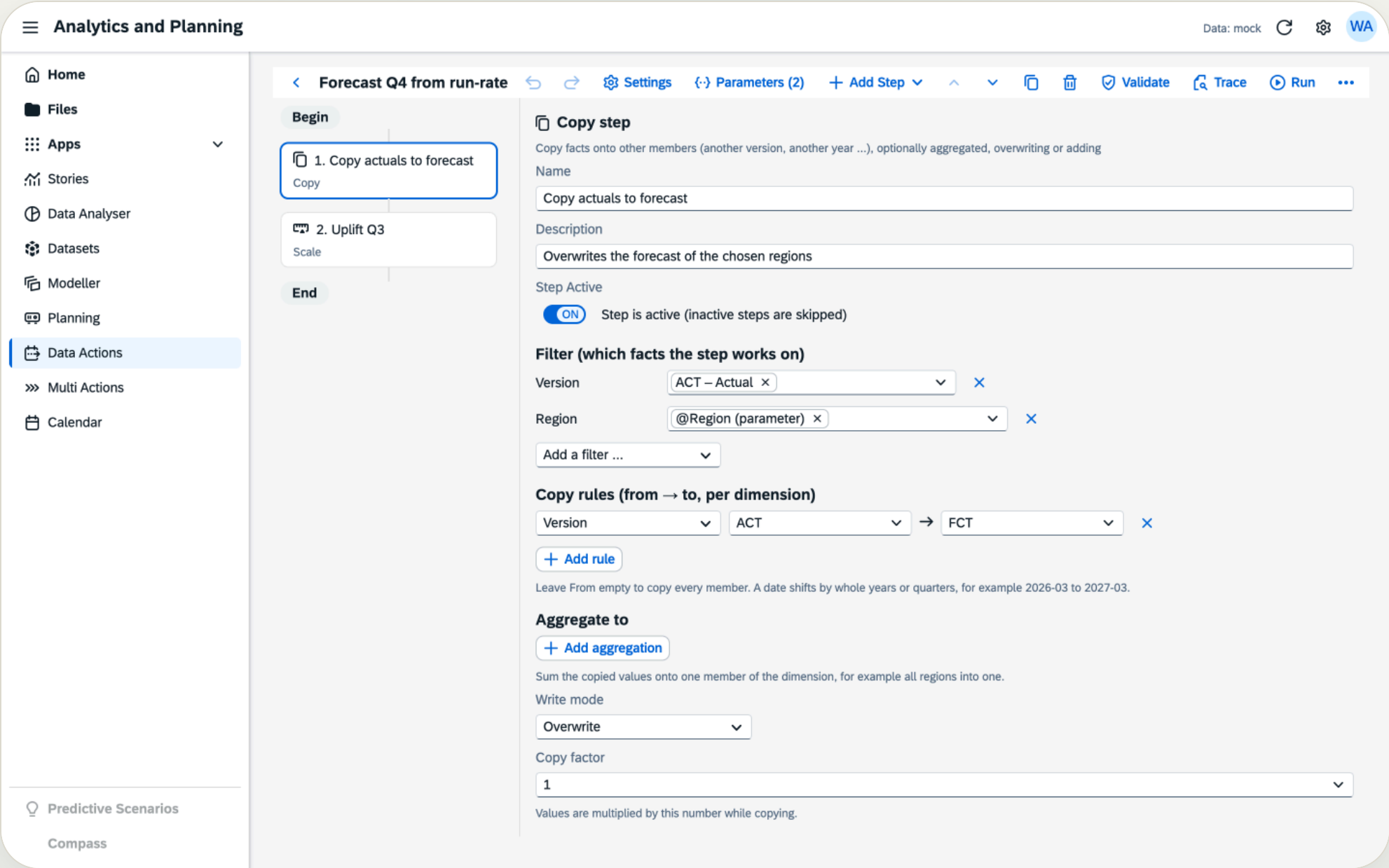
Charts, KPIs and input controls on one page, built in the designer.



SAMPLE DATA

Data actions

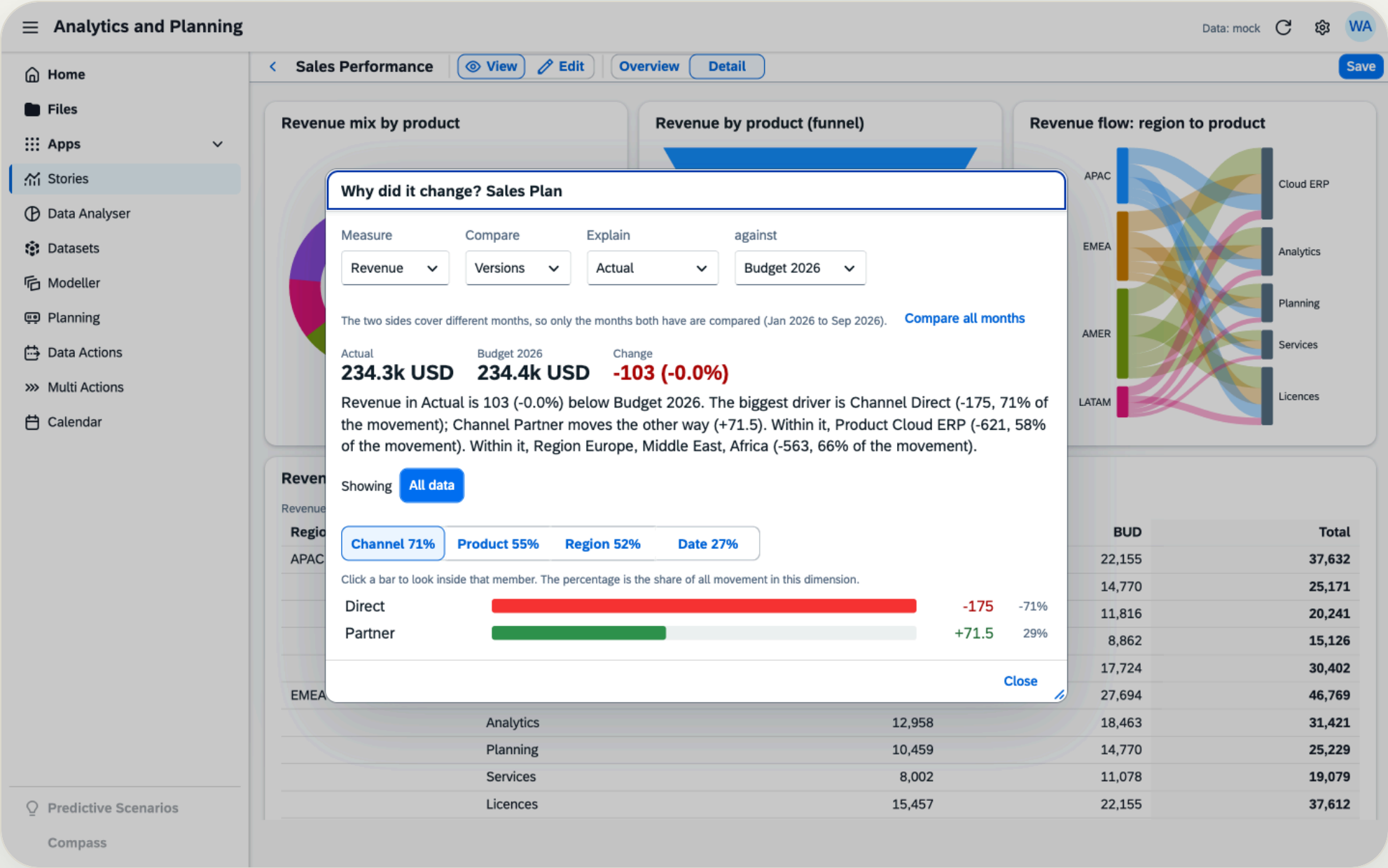
The step flow on the left, the step editor on the right. Validate, trace and run from the toolbar.



SAMPLE DATA

Why did it change?

A headline, the named drivers and ranked dimensions, with bars to drill into.



How it was tested

221 unit tests

Engines, providers, sharing, locking, calendar, charts and story layout run in Node in about a second. All pass.

28 end-to-end checks

Every page on the real system with two accounts, plus phone layout, exports and interface checks. Screenshots for each.

14 defects fixed

Found by the checks, not by users: data loss on delete, missing access rules, export noise, dark-on-dark charts, navigation.

What was tested

FEATURE	RESULT
Live CDS models, stories with filters, all widget types	Works
Planning: private version, edit, publish, revert, comments	Works
Data locking, validation rules, audit	Works
Data and multi actions; 329 facts; several currencies	Works
Sharing and security with two users	Works, after fixes
CSV and Excel export, input control filter, phone layout	Works, after fixes
Import of 1,092 values in 11 seconds	Works
PaPM step (the system has no PaPM)	As designed

Capability against SAC

CAPABILITY	THIS APP	REAL SAC
Stories, analysis, models	Wide set, eleven charts	Complete and mature
Planning with versions	Yes, with locking and rules	Plus broader workflows
Data actions	Seven step kinds, currency conversion	Plus cross-model, richer allocation
Multi actions	Eleven step types	Similar, more connectors
Value driver tree, Compass	Yes	Yes, with more options
Calendar and approvals	Yes, with Gantt and reminders	Yes, with email notices
Sharing and roles	Owner, view, edit, everyone	Teams, roles, row level rules

Where SAC goes further

CAPABILITY	THIS APP	REAL SAC
Predictive and insights	Four forecast methods, variance	Smart Predict, smart insights, Just Ask
Scripting, custom widgets	Widget registry (code)	Analytics Designer scripting
Mobile and Office	Phone layout; CSV and Excel files	Mobile app, offline, Excel add-in
Output	CSV and Excel	PDF, PowerPoint, publications, schedules
Data connections	CDS over OData V4	Many, including BW and Datasphere
Scale	Browser calculations	In-memory engine and server jobs
Transport and governance	abapGit and seed class	Content Network, transport, audit tools

COMPARED WITH SAC

Where this app is stronger

Inside the landscape

No extra product, license or tenant. It runs in S/4HANA and the Fiori launchpad, on the user's own authorizations.

Plan data stays in the ERP

No replication. Same transport and backup path, close to the source.

Open to change

Every step, widget and rule can be adapted in days. CDS sources, variance analysis and the calendar tied to planning were built for this use.

COMPARED WITH SAC

Where SAC is stronger

Wider scope

Smart Predict, Just Ask, a mobile app, the Excel add-in, scripting and PDF or PowerPoint output are not here.

Scale and speed

Calculations run in the browser. Eight steps on 1,092 rows took 85 seconds; large models will be slower.

Support and governance

We own the code and its upgrades. Plan data has no row level security, and content transport is basic.

What it would take to close the gaps

GAP TO SAC	EFFORT	APPROACH
Server side data action jobs	Large	Run the engine as an ABAP job; keep the browser run for small models
Row level security on plan data	Medium	Access control on facts by dimension member
PDF and PowerPoint output, publications	Medium	Server rendering of stories; scheduled mail
More data connections	Medium each	A source adapter per system behind the provider contract
Excel add-in	Large	Office add-in on the same OData service
Predictive and natural language	Large	Connect an AI service; the widget slots exist
Content transport	Medium	Export and import of stories and models as files

DECISION

When to use which

Use this app

Data in S/4HANA CDS views, planning with locking and rules, no second license, and a fast tailored pilot.

Use real SAC

Enterprise planning with many users and large volumes, many source systems, or PDF, Excel and mobile needs.

Use real SAC

Predictive, natural language and scripting needs, with SAP support and a product roadmap.

NEXT STEPS

Pilot on the G/L model

The feature set is complete on sample data and verified end to end on a real system with two users.

- 1 Pick the pilot team and the stories and models they share.
- 2 Measure with real volumes and decide if actions need a server job.
- 3 Choose which gaps matter first: row level security, output, connections.

Decision for Wilmar: run a pilot with one finance team on the G/L model, then decide against SAC from what the users miss.